

Jacobsthal Trees and Generalized $\kappa x \pm 1$ Transformations

Petro Kosobutskyy¹, David Mailland^{2*}

Abstract

We introduce a structural framework for the study of generalized $\kappa x \pm 1$ transformations on the set of natural numbers. The approach is based on generalized Jacobsthal sequences, which generate a family of branching structures called Jacobsthal trees.

Within this framework, classical Collatz-type iterations appear as reverse trajectories along the branches of these trees. We establish periodicity properties of the node structure, derive recurrence relations governing the formation of branching nodes, and prove structural identities linking consecutive periods.

These results provide a structural reformulation of $\kappa x \pm 1$ discrete dynamical systems in terms of Jacobsthal-type constructions.

The proposed representation reveals intrinsic periodic symmetries and partition properties of the set $\mathbb{N}$ that are not explicit in the standard iterative formulation.

Keywords: Generalized $\kappa x \pm 1$ transformations, Jacobsthal sequences, Discrete dynamical systems, Recurrence relations, Number-theoretic trees

¹ Institute of Computer Science and Information Technologies, Lviv Polytechnic National University, Lviv, Ukraine, petkosob@gmail.com 

² Independent Researcher, david.mailland@protonmail.com

*Corresponding author

Received: 5 March 2026, **Accepted:** 2 June 2026, **Available online:** 12 June 2026

How to cite this article: P. Kosobutskyy, D. Mailland, *Jacobsthal trees and generalized $\kappa x \pm 1$ transformations*, Commun. Adv. Math. Sci., 9(2) (2026), 77-91.

1. Introduction

1.1 Brief introduction and motivation

The classical Collatz problem is based on two arithmetic operations on numbers from the set $\mathbb{N}$ of natural numbers:

- Division by 2 of an even integer q_E until an odd value is reached.
- Transformation of an odd integer q_O into an even one by the Collatz function

$$C_{3q}^+ = 3q_O + 1.$$

Thus, the corresponding Collatz sequences $CS_{3,q}^+$ are conjectured to eventually reach 1 for all initial values, a property verified for numbers up to 2^{71} [1]. This naturally suggests that the classical Collatz problem admits natural generalizations. Accordingly,

the aim of this work is not to prove the classical Collatz conjecture, but to develop a generalized structural framework for $\kappa x \pm 1$ transformations based on Jacobsthal-type trees. Related studies on recurrence structures and transformations associated with generalized integer sequences include investigations on k -Narayana sequences and their binomial transforms [2], as well as works on binomial transforms of k -Jacobsthal sequences [3]. Related structural studies on generalized recursive sequence spaces were investigated in [4]. Structural approaches based on directed graph representations and acyclic graph topology were also investigated in [5]. Alternative structural and certificate-based approaches to the $3x + 1$ problem have also been proposed in recent studies on finite-certificate and 2-adic formulations [6].

In 2024, Nathanson [7] studied permutation patterns of the iterated Syracuse function (a normalised form of the classical $3x + 1$ map) and discussed its structure. Our paper extends this line of research to the general family of transformations $(1, 3, 5, 7, \dots) \cdot q \pm 1$.

The aim of this work is to establish common regularities of number transformations in the general case of the transformation function $(1, 3, 5, 7, \dots) \cdot q \pm 1$ for odd $q \in \mathbb{N}$, on the basis of which we investigate the proposed model of partitioning the set $\mathbb{N}$ into non-intersecting, isolated subsets belonging to trees of graphs with individual root sequences. As a first step, this model was originally developed by the authors [8] for the classical problem $C_{3,q}^{\pm} = 3q \pm 1$.

Here, from the perspective of transforming an odd number, mirror transformations are equivalent in structure. Indeed, since $q \in \mathbb{N}$ is odd, let $Q = -q \in \mathbb{Z}$. Then

$$3Q + 1 = -3q + 1 = -(3q - 1).$$

Hence, although the transformation $C_{3,q}^{\pm}$ is defined on the set of natural numbers $\mathbb{N}$, its extension to negative integers $\mathbb{Z}$ preserves the same sign-symmetric structure. From this perspective, both the generalized problem in the form $C_{\kappa,q}^{\pm}$ and its inverse counterpart have received only limited attention in the literature, except for the partial studies reported in [9, 10].

1.2 Preliminaries and notation

1.2.1 Generalized Jacobsthal transformations

Definition 1.1 (Generalized Jacobsthal Numbers). *The generalized Jacobsthal numbers are defined as follows:*

$$J_{\kappa,\theta,n}^{\pm} = \frac{\theta \cdot 2^n \pm (-1)^n}{\kappa}, \quad (2 \nmid \theta, n \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}).$$

Definition 1.2 (Generalized Jacobsthal Transformation). *The generalized Jacobsthal transformation is defined as follows*

$$JS_{\kappa,q}^{\pm} = \begin{cases} 2q, & q \equiv 0 \pmod{2}, \\ J_{\kappa,\theta=q,n=0}^{\pm}, & q \equiv 1 \pmod{2}, \end{cases} \quad q \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}. \quad (1.1)$$

Definition 1.3 (Residual form of the generalized Jacobsthal transformation). *The residual form of the generalized Jacobsthal transformation, equivalent to Definition 1.2, is expressed as follows:*

$$JS_{\kappa,q}^{\pm} = \begin{cases} 2q, & q \equiv 0 \pmod{2}, \\ \frac{q-1}{\kappa}, & q \equiv 1 \pmod{2}, q \equiv 1 \pmod{\kappa}, \\ \frac{q+1}{\kappa}, & q \equiv 1 \pmod{2}, q \equiv -1 \pmod{\kappa}, \\ \emptyset, & q \equiv 1 \pmod{2}, q \equiv 0 \pmod{\kappa}. \end{cases} \quad q \in \mathbb{N}_0.$$

Here, $q \equiv 0 \pmod{2}$ denotes an even number and $q \equiv 1 \pmod{2}$ an odd number.

In this context, the residual form represents the *closed-form* of the generalized Jacobsthal transformation.

Definition 1.4 (Generalized Collatz Function). *The generalized Collatz function is defined as follows*

$$C_{\kappa,q}^{+} = \kappa \cdot q + 1, \quad C_{\kappa,q}^{-} = \kappa \cdot q - 1. \quad (1.2)$$

Definition 1.5 (Generalized Collatz Conjecture). *The generalized Collatz conjecture is defined as follows*

$$C_{\kappa,q}^{\pm} = \begin{cases} \frac{q}{2}, & q \equiv 0 \pmod{2}, \\ \kappa q \pm 1, & q \equiv 1 \pmod{2}, \end{cases} \quad q \in \mathbb{N}. \quad (1.3)$$

When iterated on any initial value $q_0 \in \mathbb{N}$, the sequence eventually reaches a periodic cycle or a fixed point (Point attractor), Pas depending on κ .

1.2.2 Jacobsthal trees and nodes

The basis of the generalized problems (1.1) and (1.3) is provided by the generalized Jacobsthal numbers introduced in Definition 1.1, whose closed form was established in earlier works [9, 10].

They can be structured into two subclasses of integers:

$$m_{\kappa,\theta,n} = \frac{\theta \cdot 2^{n-1} - 1}{\kappa}, \quad p_{\kappa,\theta,k} = \frac{\theta \cdot 2^{k-1} + 1}{\kappa}, \quad n \in \mathbb{N}_{\text{even}}, k \in \mathbb{N}_{\text{odd}}. \quad (1.4)$$

Whenever these quantities are integers, they generate a new odd parameter thus defining the branching nodes of the corresponding Jacobsthal tree.

$$\delta \in \mathbb{N}_{\text{odd}}, \quad \delta = \begin{cases} m_{\kappa,\theta,n}, & \text{if } \theta \equiv 1 \pmod{\kappa} \quad (JS^+), \\ p_{\kappa,\theta,k}, & \text{if } \theta \equiv -1 \pmod{\kappa} \quad (JS^-). \end{cases}$$

For odd θ and indices $n \in \mathbb{N}_{\text{even}}, k \in \mathbb{N}_{\text{odd}}$,

$$\begin{aligned} m_{\kappa,\theta,n} \in \mathbb{N}_0 &\iff \theta 2^n \equiv 1 \pmod{\kappa}, \\ p_{\kappa,\theta,k} \in \mathbb{N}_0 &\iff \theta 2^k \equiv -1 \pmod{\kappa}, \end{aligned}$$

The Jacobsthal numbers themselves were studied in [11, 12], and the authors pointed out their possible role in the Collatz problem. However, as shown in [9, 10], the numbers (1.4) form, at certain points of sequences, the so-called *branching nodes* of other sequences

$$\{\theta \cdot 2^n\}_{n=0}^{+\infty}$$

based on powers of two. Such a process develops as $n \rightarrow +\infty$, resulting in the formation of a so-called *graph tree*, as in [13]. In the inverse iteration direction, along the branches of the tree, Collatz sequences $(CS_{\kappa,q}^{\pm})$ are formed.

Definition 1.6 (Point attractor). *A point attractor (PA) is an attractor of a dynamical system that consists of a single stationary point.*

In the periodic cycle that terminates a Collatz sequence $CS_{\kappa,q}^{\pm}$, this role PA is played by the smallest odd value q_0 . In a non-periodic cycle with increasing sequences, the smallest odd value $q_0 = 1$ is a pseudo-attractor (PA_{ps}).

Therefore, the corresponding generalized Collatz conjecture asserts that, for any initial value $q_0 \in \mathbb{N}_{PA(PA_{ps})}$, the iterated sequence defined by

$$q_{n+1} = C_{\kappa,q_0}^{\pm}$$

eventually reaches the set $PA(PA_{ps})$.

Remark 1.7. *In the classical Collatz model with $\kappa = 3$ and $C(x) = 3x + 1$, the periodic attractor is $PA = \{1\}$ for all $q \in \mathbb{N}$. In the case of $C(x) = 3x - 1$, one obtains $PA = \{1, 5, 17\}$; for $C(x) = 5x + 1$, $PA = \{1, 13, 17\}$; and for $C(x) = 181x + 1$, $PA = \{1, 27, 35\}$.*

Definition 1.8 (Node). A node is a point of the sequence

$$\{\theta 2^n\}_{n=0}^{+\infty}$$

that corresponds to the numbers $m_{\kappa,\theta,n}$ or $p_{\kappa,\theta,n}$ defined in (1.4).

The points of sequences $\{\theta \cdot 2^n\}_{n=0}^{+\infty}$ with numbers $m_{\kappa,\theta,n}$ or $p_{\kappa,\theta,n}$ correspond to the so-called nodes, from which other sequences branch, if the parameter θ is not a multiple of κ . If θ is a multiple of κ ($\theta = \theta_\kappa$), then the numbers $m_{\kappa,\theta,n}$ (or $p_{\kappa,\theta,n}$) are not integers, so no branching nodes are generated.

An illustration of nodes with $m_{\kappa,\theta,n}$ numbers is shown in Figure 1.1.

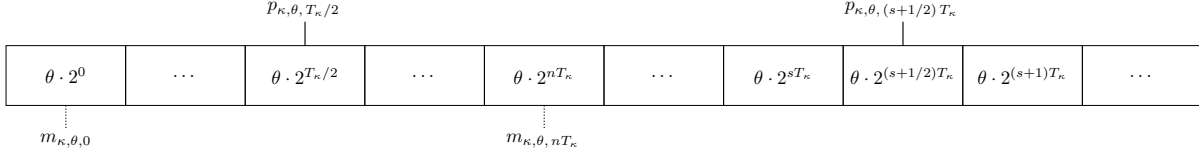


Figure 1.1. Arrangement of the numbers $m_{\kappa,\theta,n}$ (even indices, below) and $p_{\kappa,\theta,k}$ (odd indices, above) along the sequence $\theta \cdot 2^n$. If the values $m_{\kappa,\theta,n}$ and $p_{\kappa,\theta,k}$ are not integers (see Eq. (1.4) and Table 2.2), no nodes are defined.

Definition 1.9 (Period of the Nodes). The period of the nodes T_κ in the sequence

$$\{\theta 2^n\}_{n=0}^{+\infty}$$

is the distance, in powers of 2^n , between two successive nodes of the same type.

Definition 1.10 (Jacobsthal tree). A Jacobsthal tree is the branching structure generated by the graph nodes of Jacobsthal sequences along increasing powers of two.

Definition 1.11 (Sets of numbers associated with the Jacobsthal tree). The sets associated with the Jacobsthal tree are defined as follows.

The graph tree rooted at the point attractor PA corresponds to the combined set consisting of odd numbers $N_{\kappa,PA,O}$ as nodes and even numbers $N_{\kappa,PA,E}$ as edges:

$$N_{\kappa,PA} = \bigcup_{i=1}^k N_{\kappa,PA,i} \cup \bigcup_{i=1}^k N_{\kappa,E,i}.$$

Definition 1.12 (Total stopping time (tst)). We define the total stopping time tst as follows [14].

In the Jacobsthal tree, the discrete time variable tst corresponds to the number of iterations ($n = tst$). Therefore, in the forward direction $n \rightarrow +\infty$, the time tst increases, while in the reverse construction of the Collatz sequence, tst decreases.

1.2.3 Structural interpretation

Informally, the proposed framework may be viewed as follows: Starting from root sequences of the form $\theta 2^n$, certain values generate integer branching nodes through the generalized Jacobsthal relations. These nodes form tree-like structures, called Jacobsthal trees, whose reverse trajectories correspond to generalized Collatz-type sequences. In this representation, periodicity properties of the nodes induce structural regularities in the associated $\kappa x \pm 1$ dynamics.

2. Main Results

2.1 Generalized Collatz and Jacobsthal transformations

In the forward direction ($n \rightarrow +\infty$), at nodes of the form $\{\delta 2^n\}_{n \geq 0}$ with $\delta \in \{m_{\kappa,\theta,n}, p_{\kappa,\theta,n}\}$ whenever they are integers, other sequences branch.

In the reverse direction ($n \rightarrow 0$), using the relations

$$\kappa m_{\kappa,\theta,n} + 1 = \theta 2^n, \quad \kappa p_{\kappa,\theta,k} - 1 = \theta 2^k, \quad n \in \mathbb{N}_{\text{even}}, \quad k \in \mathbb{N}_{\text{odd}} \quad (2.1)$$

the segments $\{\delta 2^n\}_{n=1}^\ell$ merge at the nodes, forming Collatz-type sequences $CS_{\kappa,q}^\pm$.

Therefore, for every κ , the Collatz functions can be expressed as

$$C_{\kappa,q}^+ = \kappa m_{\kappa,\theta,n} + 1, \quad C_{\kappa,q}^- = \kappa p_{\kappa,\theta,k} - 1. \quad (2.2)$$

A similar reasoning applies to the system of nodes with numbers $p_{\kappa,\theta,k}$, leading to the mirror counterpart Collatz function $C_{\kappa,q}^-$.

In this case,

$$p_{\kappa,\theta,k} = \frac{\theta 2^k + 1}{\kappa} \iff \theta 2^k = \kappa p_{\kappa,\theta,k} - 1, \quad C_{\kappa,q}^- = \kappa p_{\kappa,\theta,k} - 1. \quad (2.3)$$

In justifying formulas (2.1)–(2.3), no additional constraints were imposed; therefore, both models $C_{\kappa,q}^\pm$ are structurally equivalent for all κ .

Thus, the Collatz-type problem can be generalized as

$$C_{\kappa,q}^\pm = \begin{cases} \frac{q}{2}, & \text{if } q \equiv 0 \pmod{2}, \\ \kappa \cdot q \pm 1, & \text{if } q \equiv 1 \pmod{2}, \end{cases} \quad n \rightarrow 0, \quad (2.4)$$

and the reverse (Jacobsthal-type) problem as

$$JS_{\kappa,q}^\mp = \begin{cases} 2q, & q \equiv 0 \pmod{2}, \\ \frac{q \pm 1}{\kappa}, & q \equiv 1 \pmod{2}, \end{cases} \quad n \rightarrow +\infty. \quad (2.5)$$

Since the graph tree serves as the basis for forming Collatz sequences, problem (2.4) should be considered as the direct problem, and (2.5) as the inverse problem of number transformation.

2.2 Examples of Jacobsthal trees

At the end of the article, an alternative formulation of a Collatz-type hypothesis is proposed.

To identify the transformed nodes, we distinguish two families:

- Nodes of type $m_{3,q,n}$ (for JS^+), defined as

$$m_{3,q,n} = \frac{q \cdot 2^n - 1}{3}, \quad \text{defined whenever the numerator is divisible by 3.}$$

- Nodes of type $p_{3,q,n}$ (for JS^-), defined as

$$p_{3,q,n} = \frac{q \cdot 2^n + 1}{3}, \quad \text{defined whenever the numerator is divisible by 3.}$$

Thus, we evaluate $J_{3,q,n}^\pm$ for q in the range 1 to 9. For $q = 1$, the corresponding values are shown in Table 2.1.

Therefore, in model (1.4), the numbers $J_{3,1,n}^-$, unlike $J_{3,1,n}^+$, are divisible by 3, 9, 15, 21, ... However, both numbers $J_{3,1,n}^\pm$ are divisible by 5. Thus, $J_{3,1,n}^+$ generate numbers $p_{3,1,n}$, while $J_{3,1,n}^-$ generate numbers $m_{3,1,n}$, thereby ensuring non-overlap of the subsets of numbers $m_{3,1,n}$ and $p_{3,1,n}$.

The numbers $J_{3,1,n}^+$ are not integers, whereas $J_{3,1,n}^- = J_{1,1,n}^-/3$ are integers. Therefore, numbers $m_{3,1,n}$ and $p_{3,1,n}$ are obtained by bisection of $J_{3,1,n}^\pm$, thereby guaranteeing that the sequences $CS_{3,q}^\pm$ terminate in unity.

For $\kappa > 1$, the numbers $J_{\kappa,1,n}^\pm$ are generated differently. For example, if

$$m_{5,1,n} = \frac{J_{1,1,n}^-}{5}, \quad \text{and} \quad p_{5,1,n} = \frac{J_{1,1,n}^+}{5},$$

then

$$m_{7,1,n} = \frac{J_{1,1,n}^-}{7}$$

are generated by uniting the sequences $J_{7,1,n}^+ \cup J_{7,1,n}^-$, whereas the numbers $p_{7,1,n}$ are not integers. The case $\kappa = 9$ is analogous to $\kappa = 3$.

n	0	1	2	3	4	5	6	7	8	9	10	11	12	
$J_{1,1,n}^+$	2	1	5	7	17	31	65	127	257	511	1025	2047	4097	$p_{1,1,n} = 2n - 1$
$J_{1,1,n}^-$	0	3	3	9	15	33	63	129	255	513	1023	2049	4095	$m_{1,1,n} = 2n + 1$
$J_{3,1,n}^- = J_{1,1,n}^-/3$	0	1	1	3	5	11	21	43	85	171	341	683	1365	$m(p)_{3,1,n}$
$J_{5,1,n}^+ = J_{1,1,n}^+/5$			1				13				205			$p_{5,1,n}$
$J_{5,1,n}^- = J_{1,1,n}^-/5$	0				3				51				819	$m_{5,1,n}$
$J_{7,1,n}^+ = J_{1,1,n}^+/7$				1						73				
$J_{7,1,n}^- = J_{1,1,n}^-/7$	0						9						585	$m_{7,1,n}$
$J_{7,1,n}^- \cup J_{7,1,n}^+$	0			1			9			73			585	
$J_{9,1,n}^- = J_{1,1,n}^-/9$	0			1			7			57			455	$m(p)_{9,1,n}$

Table 2.1. Computed values of $J_{3,q,n}^\pm$ for the transformation model $C_{3,q}^\pm$ within the range $q = 1, \dots, 9$. Highlighted rows in yellow indicate subsequences corresponding to $q = 1, 3, 5, 7, 9$. The rightmost column provides the associated generating expressions $p_{3,1,n}$ and $m_{3,1,n}$, defining the mappings $q \mapsto 3q \pm 1$.

If $q > 1$, divisibility by three among the numbers $J_{1,q,n}^\pm$ is selective: certain numbers $J_{1,3,n}^\pm$ are not divisible by three, so the non-integer numbers are $p_{3,3,n}$ and $m_{3,3,n}$, as are the numbers $p_{\kappa,q=\kappa,n}$ and $m_{\kappa,q=\kappa,n}$.

The correlation of numbers $m_{\kappa,q,n}$ and $p_{\kappa,q,n}$ with $J_{\kappa,1,n}^\pm$ is displayed in the right column of Table 2.1.

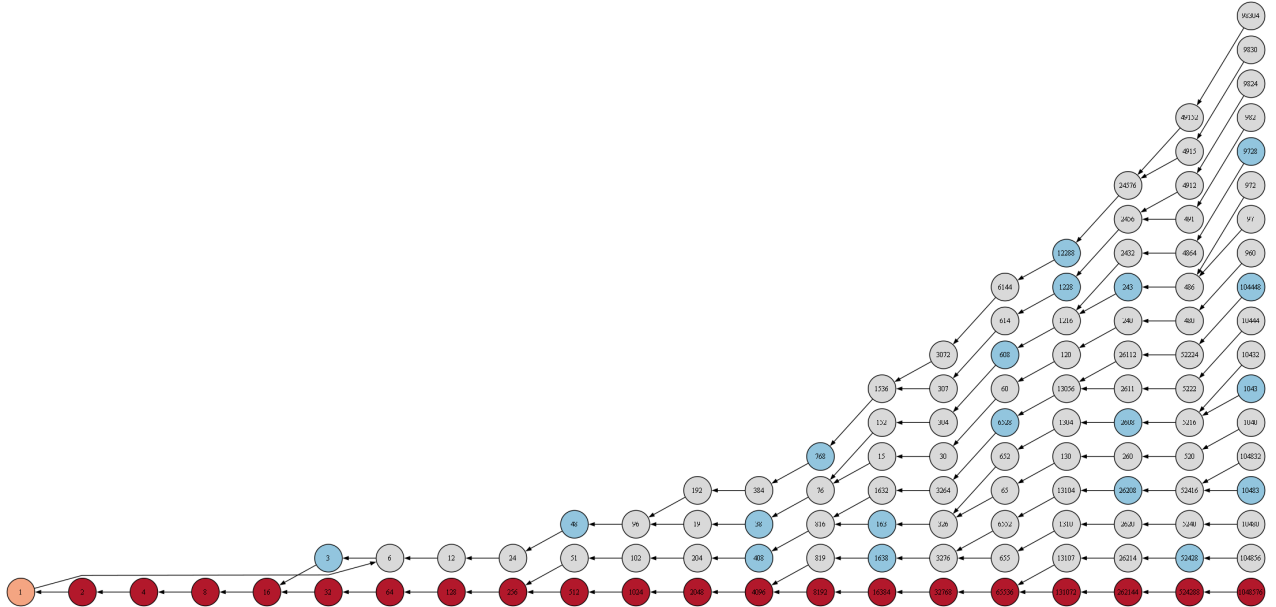


Figure 2.1. Representation of the Jacobsthal tree $J_{5,q}^+$, branching as $n \rightarrow +\infty$. In the reverse direction, the arrows indicate the trajectories of the Collatz sequences $C_{5,q}^+$ formed from the initial numbers in the columns.

2.3 Structural interpretation of the trees

Transformations (2.4) and (2.5) are inverse to each other in such a way that, as $n \rightarrow +\infty$ of graph (2.5), branching occurs in the form of a tree (the Jacobsthal tree), as shown in Figure 2.1 for the $J_{5,q}^+$ tree.

In the reverse direction $n \rightarrow 0$, sequences $CS_{5,q}^+$ are formed that converge along the branches of the tree toward the root

$$\{2^n\}_{n=0}^{+\infty}$$

sequence. The sequences $CS_{5,q}^+$ with initial numbers belonging to one column have the same total stopping time and the same tst time. These regularities hold true for other models of κ -conjectures. Therefore, as $tst \rightarrow +\infty$, the tree formation process described below is unambiguous, since each value of tst corresponds to a definite number and a list of numbers in the columns, for which the sequences $CS_{\kappa,q}^\pm$ have the same length. The inverse mapping is non-unique, since for the same number of iterations (or time tst), the unit value is reached by the sequences $CS_{\kappa,q}^\pm$ of numbers in the entire column.

Since the set of initial numbers is fixed/defined, their sequences $CS_{\kappa,q}^\pm$ reach the attractor (or pseudo-attractor) in the same number of iterations.

2.4 Periodicity theorem

We now formulate the main structural results of problems (2.4) and (2.5).

Theorem 2.1. *For all κ , the nodes located at points of the sequences $\{\theta \cdot 2^n\}_{n=0}^{+\infty}$ are periodic with period T_κ with respect to powers of two, while the numbers $m_{\kappa,\theta,n}$ and $p_{\kappa,\theta,n}$, as well as those of the neighbouring nodes, satisfy the linear relations*

$$2^{T_\kappa} = 1 + \kappa m_{\kappa,1,T_\kappa}, \quad (2.6a)$$

$$m_{\kappa,\theta,n+T_\kappa} = 2^{T_\kappa} m_{\kappa,\theta,n} + m_{\kappa,1,T_\kappa}, \quad (2.6b)$$

$$p_{\kappa,\theta,n+T_\kappa} = 2^{T_\kappa} p_{\kappa,\theta,n} - m_{\kappa,1,T_\kappa}, \quad (2.6c)$$

where $m_{\kappa,1,T_\kappa}$ denotes the first odd number of the root sequence $\{1 \cdot 2^n\}_{n=0}^{+\infty}$.

2.5 Periodic node formation

We verify formula (2.6a) as follows:

κ	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	
$m_{\kappa,1,n}$	1	1	3	1	7	93	315	1	15	13 797	3	89	41 943	9 709	9 256 395	
$p_{\kappa,1,k}$				f				f			f	f	f			
T_κ	1	2	4	3	6	10	12	4	8	18	6	11	20	18	28	
κ	31	33	181													
$m_{\kappa,1,n}$	1	31	8 466 826 192 629 220 211 924 569 210 775 188 859 772 039 349 080 675													
$p_{\kappa,1,k}$	f		f													
T_κ	5	10	180													

Table 2.2. Values of $m_{\kappa,1,T_\kappa}$ and periods T_κ for selected values of κ .

The gray-shaded blocks indicate triplets of parameters κ , $\kappa + 2$, and $m_{\kappa+2,1,T_\kappa}$ satisfying

$$\kappa = m_{\kappa+2,1,T_\kappa} \quad \text{and} \quad T_\kappa = \frac{T_{\kappa+2}}{2}.$$

From (2.6), if the period T_κ is odd, as in the cases $\kappa = 7, 23, 31, 47, 49, \dots$, then the quantities $p_{\kappa,\theta,n}$ are not integers.

However, numbers $p_{\kappa,\theta,n}$ can take non-integer values for even period values as in the cases $\kappa = 35, 39, 45, 51, \dots$

Remark 2.2. *From formula (2.6a) we have $2^{T_\kappa} - 1 = \kappa \cdot m_{\kappa,1,T_\kappa}$. Therefore, the factorization of $2^{T_\kappa} - 1$ can be expressed in the form of binary products of the type $\kappa \cdot m_{\kappa,1,T_\kappa}$.*

We now justify formula (2.6b). From Figure 2.2(a), the node numbers $p_{\kappa,\theta,n}$ and $m_{\kappa,\theta,n}$ are used to denote the two corresponding families of sequence elements.

The transformations for both can be written as:

$$\theta \cdot 2^n = \kappa \cdot p_{\kappa,\theta,n} - 1, \quad \theta \cdot 2^{n+T_\kappa} = \kappa \cdot p_{\kappa,\theta,n+T_\kappa} - 1.$$

Therefore, for the odd numbers:

$$\kappa \cdot p_{\kappa, \theta, n} - 1 = \frac{\kappa \cdot p_{\kappa, \theta, n+T_\kappa} - 1}{2^{T_\kappa}},$$

or equivalently:

$$p_{\kappa, \theta, n+T_\kappa} = 2^{T_\kappa} p_{\kappa, \theta, n} - m_{\kappa, 1, T_\kappa}.$$

For the even numbers, the analogous relation is:

$$\theta \cdot 2^n = \kappa \cdot m_{\kappa, \theta, n} + 1, \quad \theta \cdot 2^{n+T_\kappa} = \kappa \cdot m_{\kappa, \theta, n+T_\kappa} + 1,$$

leading to

$$m_{\kappa, \theta, n+T_\kappa} = 2^{T_\kappa} m_{\kappa, \theta, n} + m_{\kappa, 1, T_\kappa}.$$

Here, $p_{\kappa, 1, T_\kappa}$ and $m_{\kappa, 1, T_\kappa}$ denote the first odd and even nodes of the corresponding root sequences $\{1 \cdot 2^n\}_{n=0}^{n=+\infty}$.

We present the analytical forms of formulas (2.6b) within the range of values $\kappa = 1$ to 9.

κ	1	3	5	7	9	181
$m_{\kappa, \theta, n+T_\kappa}$	$2^1 m_{1, \theta, n} + 1$	$2^2 m_{3, \theta, n} + 1$	$2^4 m_{5, \theta, n} + 3$	$2^3 m_{7, \theta, n} + 1$	$2^6 m_{9, \theta, n} + 7$	$2^{180} m_{181, \theta, n}$ + 84 668 261 926 292 202 11 924 569 210 775 188 859 772 039 349 080 675

Table 2.3. Examples of recurrence relations of type (2.6b) for selected values of κ .

2^n	$m_{1,1,n}$	$m_{3,1,n}$	$m_{5,1,n}$	$m_{7,1,n}$	$m_{9,1,n}$	$m_{11,1,n}$	$m_{13,1,n}$	$m_{15,1,n}$	$m_{17,1,n}$	$m_{19,1,n}$	$m_{21,1,n}$	$m_{23,1,n}$
2^{20}	1 048 575	349 525	29 715			95 325		69 905				
2^{19}	524 287											
2^{18}	262 143	87 381		37 449	29 127					13 797	12 483	
2^{17}	13 107											
2^{16}	65 535	21 845	13 107					4 369	3 855			
2^{15}	32 767			4 681								
2^{14}	16 383	5 461										
2^{13}	8 191											
2^{12}	4 095	1 365	819	585	455		315	273			195	
2^{11}	2 045											89
2^{10}	1 023	341				93						
2^9	511			73								
2^8	255	85	51					17	15			
2^7	127											
2^6	63	21		9	7						3	
2^5	31											
2^4	15	5	3					1				
2^3	7			1								
2^2	3	1										
2^1	1											
2^0	0	0	0	0	0	0	0	0	0	0	0	0

Table 2.4. Illustration of the node formation property with the numbers $m_{\kappa, 1, n}$ for $\kappa = 1$ to $\kappa = 9$.

The periodicity of node formation with numbers $m_{\kappa, \theta, n}$ is shown in Table 2.4, where the values of the first odd numbers $m_{\kappa, 1, T_\kappa}$ in formula (2.6a) are highlighted in yellow.

The period T_κ also correlates with the total stopping time tst , as shown in Table 2.5 for the transformation function $C_{5,q}^+(T_5 = 4)$.

$m_{5,1,n}$	0	3	51	819	13 107	209 715
$tst + T_5$	1	1 + 4	5 + 4 = 9	9 + 4 = 13	13 + 4 = 17	17 + 4 = 21
		5	9	13	17	21

Table 2.5. Example of computed values for $m_{5,1,n}$ and increments by T_5 .

Taking into account the structure of the nodes in Figure 1.1, for the numbers $p_{\kappa,\theta,n+T_\kappa}$ as shown in Table 2.3, the results are analogous, except with a minus sign.

In (2.6b), the first formula

$$m_{3,\theta,n+2} = 4m_{3,\theta,n} + 1$$

is known [15].

The corresponding relations for other values of κ were previously investigated in [9].

Formulas of type (2.6b) are justified as $n \rightarrow +\infty$, i.e., they reflect regularities in the transformation of parameterized θ odd Jacobsthal numbers $m_{\kappa,\theta,n}$ and $p_{\kappa,\theta,n}$.

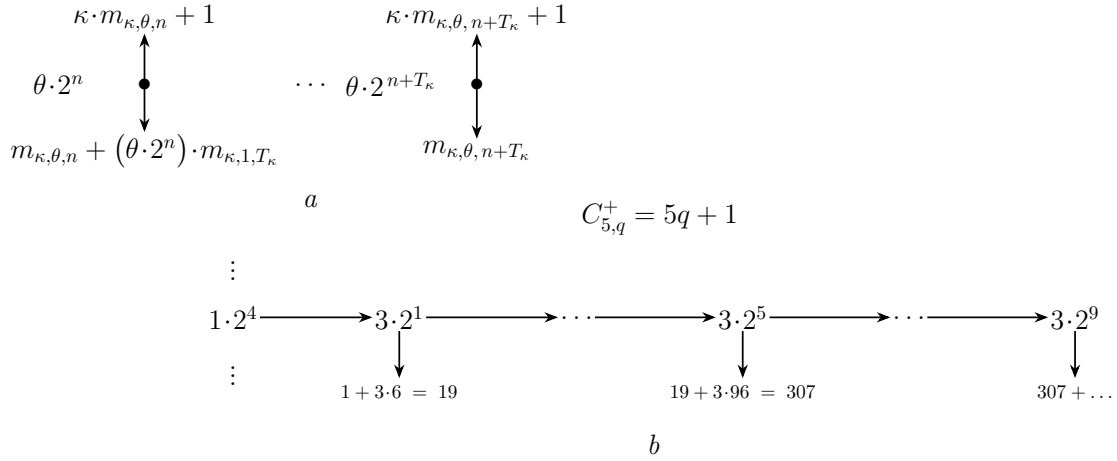


Figure 2.2. (a) Generalized graph associated with (2.4) and (2.5); (b) Graph associated with (2.5).

2.6 Structural properties of node transformations

Property 2.3. For all κ , the difference between two numbers that are adjacent in period T_κ is proportional to the transformation function of odd numbers $C_{\kappa,q}^\pm$:

$$\begin{cases} m_{\kappa,\theta,n+T_\kappa} - m_{\kappa,\theta,n} = m_{\kappa,1,T_\kappa} (\kappa m_{\kappa,\theta,n} + 1) = m_{\kappa,1,T_\kappa} C_{\kappa,q}^+, \\ p_{\kappa,\theta,n+T_\kappa} - p_{\kappa,\theta,n} = m_{\kappa,1,T_\kappa} (\kappa p_{\kappa,\theta,n} - 1) = m_{\kappa,1,T_\kappa} C_{\kappa,q}^-. \end{cases}$$

Proof. From (2.6b), we have

$$m_{\kappa,\theta,n+T_\kappa} - m_{\kappa,\theta,n} = (2^{T_\kappa} - 1)m_{\kappa,\theta,n} + m_{\kappa,1,T_\kappa}.$$

Using (2.6a), namely

$$2^{T_\kappa} - 1 = \kappa m_{\kappa,1,T_\kappa},$$

we obtain

$$m_{\kappa,\theta,n+T_\kappa} - m_{\kappa,\theta,n} = m_{\kappa,1,T_\kappa} (\kappa m_{\kappa,\theta,n} + 1) = m_{\kappa,1,T_\kappa} C_{\kappa,q}^+.$$

Similarly, from (2.6c),

$$p_{\kappa, \theta, n+T_\kappa} - p_{\kappa, \theta, n} = (2^{T_\kappa} - 1)p_{\kappa, \theta, n} - m_{\kappa, 1, T_\kappa}.$$

Using again (2.6a), we obtain

$$p_{\kappa, \theta, n+T_\kappa} - p_{\kappa, \theta, n} = m_{\kappa, 1, T_\kappa} (\kappa p_{\kappa, \theta, n} - 1) = m_{\kappa, 1, T_\kappa} C_{\kappa, q}^-.$$

□

Property 2.4. For any κ , the following identities hold:

$$\begin{cases} m_{\kappa, \theta, n} + p_{\kappa, \theta, n+T_\kappa/2} = \frac{\theta 2^n}{\kappa} (1 + 2^{T_\kappa/2}), & \text{if the numbers } p_{\kappa, \theta, n} \text{ are integers,} \\ -m_{\kappa, \theta, n} + m_{\kappa, \theta, n+T_\kappa} = \theta 2^n m_{\kappa, 1, T_\kappa}, & \text{if the numbers } p_{\kappa, \theta, n+T_\kappa} \text{ are not integers.} \end{cases}$$

Proof. From (2.1), we have:

$$m_{\kappa, \theta, n} + p_{\kappa, \theta, n+T_\kappa/2} = \frac{\theta \cdot 2^n - 1}{\kappa} + \frac{\theta \cdot 2^n \cdot 2^{T_\kappa/2} + 1}{\kappa} = \frac{\theta \cdot 2^n}{\kappa} (1 + 2^{T_\kappa/2}),$$

and

$$-m_{\kappa, \theta, n} + m_{\kappa, \theta, n+T_\kappa} = \frac{-\theta \cdot 2^n + 1}{\kappa} + \frac{\theta \cdot 2^n \cdot 2^{T_\kappa} - 1}{\kappa} = \frac{\theta \cdot 2^n}{\kappa} (-1 + 2^{T_\kappa}) = \theta 2^n m_{\kappa, 1, T_\kappa}.$$

□

Property 2.5. For all κ , the asymptotic ratios satisfy

$$\begin{aligned} \lim_{n \rightarrow +\infty} \frac{m_{\kappa, \theta, n+T_\kappa}}{m_{\kappa, \theta, n}} &= 2^{T_\kappa}, \\ \lim_{n \rightarrow +\infty} \frac{p_{\kappa, \theta, n+T_\kappa}}{p_{\kappa, \theta, n}} &= 2^{T_\kappa}. \end{aligned}$$

Proof. From the periodicity relations (2.6b) and (2.6c), we have

$$m_{\kappa, \theta, n+T_\kappa} = 2^{T_\kappa} m_{\kappa, \theta, n} + m_{\kappa, 1, T_\kappa},$$

and

$$p_{\kappa, \theta, n+T_\kappa} = 2^{T_\kappa} p_{\kappa, \theta, n} - m_{\kappa, 1, T_\kappa}.$$

Dividing the first identity by $m_{\kappa, \theta, n}$ gives

$$\frac{m_{\kappa, \theta, n+T_\kappa}}{m_{\kappa, \theta, n}} = 2^{T_\kappa} + \frac{m_{\kappa, 1, T_\kappa}}{m_{\kappa, \theta, n}}.$$

Since $m_{\kappa, \theta, n} \rightarrow +\infty$ as $n \rightarrow +\infty$, the second term tends to zero. Hence,

$$\lim_{n \rightarrow +\infty} \frac{m_{\kappa, \theta, n+T_\kappa}}{m_{\kappa, \theta, n}} = 2^{T_\kappa}.$$

Similarly, dividing the second identity by $p_{\kappa, \theta, n}$ gives

$$\frac{p_{\kappa, \theta, n+T_\kappa}}{p_{\kappa, \theta, n}} = 2^{T_\kappa} - \frac{m_{\kappa, 1, T_\kappa}}{p_{\kappa, \theta, n}}.$$

Since $p_{\kappa, \theta, n} \rightarrow +\infty$ as $n \rightarrow +\infty$, the second term also tends to zero. Therefore,

$$\lim_{n \rightarrow +\infty} \frac{p_{\kappa, \theta, n+T_\kappa}}{p_{\kappa, \theta, n}} = 2^{T_\kappa}.$$

□

2.7 Partition properties

Property 2.6. For a given κ we have:

$$\text{If } \theta \neq \delta, \text{ then } \begin{cases} m_{\kappa, \theta, n} \neq m_{\kappa, \delta, v}, \\ p_{\kappa, \theta, n} \neq p_{\kappa, \delta, v}. \end{cases}$$

Proof. Assume that

$$m_{\kappa, \theta, n} = m_{\kappa, \delta, v}.$$

Then, from (2.1),

$$\theta \cdot 2^n = \delta \cdot 2^v,$$

which implies

$$\frac{\theta}{\delta} = 2^{v-n}.$$

Since both θ and δ are odd, the ratio θ/δ cannot be a nontrivial power of two. Therefore,

$$\theta = \delta \quad \text{and} \quad n = v,$$

which contradicts the assumption $\theta \neq \delta$. Hence,

$$m_{\kappa, \theta, n} \neq m_{\kappa, \delta, v}.$$

The proof for the family $p_{\kappa, \theta, n}$ is analogous.

Therefore, the sets $S(m_{\kappa, \theta, n})$ and similarly $S(p_{\kappa, \theta, n})$, contain no identical elements.

□

2.8 Closed cycles and pseudo-attractors

In the classical Collatz sequence, the sequence $CS_{3,q}^+$ converges to a periodic cycle with a unit smallest odd value ($q_o = 1$). Conjectures $C_{3,q_o}^- = 3q_o - 1$, $C_{5,q_o}^+ = 5q_o + 1$, $C_{181,q_o}^+ = 181q_o + 1$, are known [16–21], in which the sequence $CS_{\kappa,q}^\pm$ converges to a periodic cycle with minimal $q_o > 1$, or grows without bound if $q_o \geq 5$, as shown in Table 2.6.

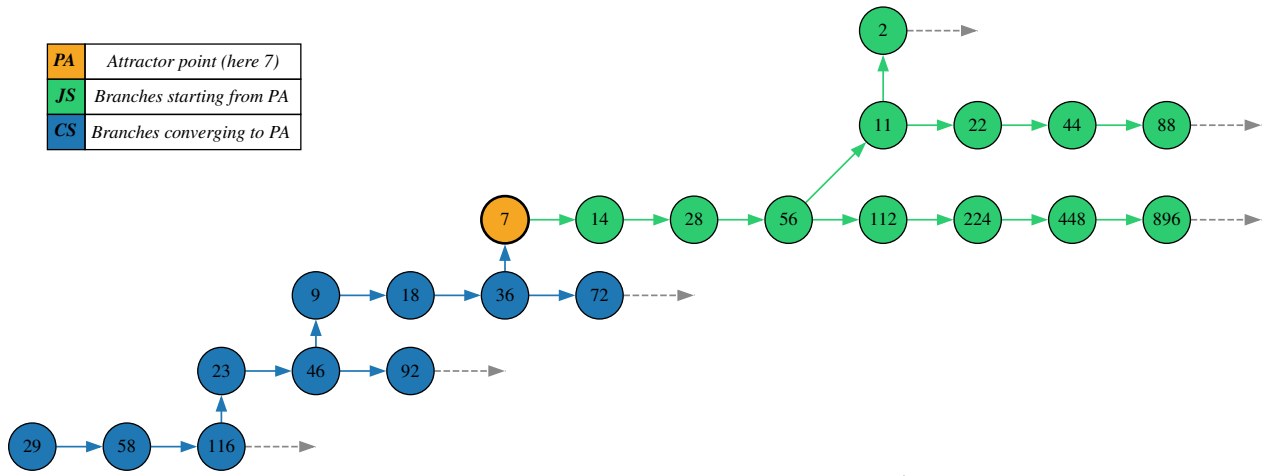


Figure 2.3. The graphs of the sequences $JS_{5,q}^-$ (in the arrow direction) and $CS_{5,q}^+$ (in the reverse direction).

Cases with divergent trajectories correspond to increasing sequences $CS_{\kappa,q}^\pm$.

κ	$C_{\kappa,q_0}^{\pm}$	PA	Type of cycle	$PA_{\kappa} = \sum_{i=0}^{k-1} \frac{\kappa^{k-v-1} 2^v}{2^{\ell} - \kappa^k}$	$\mathbb{N}_{(\pm, \kappa, PA)}$	$\mathbb{N}_{(\pm, \kappa, PA)} / \mathbb{N}$
1	C_{1,q_0}^{+}	1	$\rightarrow 1 \rightarrow 0$	$PA_1 = 1/(3^1 \cdot 2^1) = 1$	$\mathbb{N}_{(+,1,1)}$	—
	C_{1,q_0}^{-}	1	$\rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow \dots$	$PA_1 = 1/(3^1 - 2^1) = 1$	$\mathbb{N}_{(-,1,1)}$	—
3	C_{3,q_0}^{+}	1	$\rightarrow 1 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow \dots$	$PA_3 = 1/(2^2 - 3^1) = 1$	$\mathbb{N}_{(+,3,1)}$	1
	$C_{3,q}^{-}$	1	$\rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow \dots$	$PA_3 = 1/(3^1 - 2^2) = 1$	$\mathbb{N}_{(-,3,1)}$	32.77%
		5	$\dots \rightarrow 5 \rightarrow 7 \rightarrow 5 \rightarrow \dots$	$PA_3 = \frac{3^1 + 2^1}{(3^2 - 2^3)} = 5$	$\mathbb{N}_{(-,3,5)}$	32.34%
		17	$\rightarrow 17 \rightarrow 25 \rightarrow 37 \rightarrow 55 \rightarrow 41 \rightarrow 61 \rightarrow 91 \rightarrow 17 \rightarrow \dots$	$PA_3 = \frac{3^6 + \dots + 2^{11}}{3^7 - 2^{11}} = \frac{2363}{139} = 17$	$\mathbb{N}_{(-,3,17)}$	34.90%
5	C_{5,q_0}^{-}	1	$\dots \rightarrow 1 \rightarrow 1 \rightarrow \dots$	$PA_5 = 1/(5^1 - 2^2) = 1$	$\mathbb{N}_{(-,5,1)}$	—
		∞	$\rightarrow 1 \rightarrow \infty \rightarrow \dots$	—	$\mathbb{N}_{(-,5,\infty)}$	—
	C_{5,q_0}^{+}	1	$\dots \rightarrow 1 \rightarrow 3 \rightarrow 1 \rightarrow \dots$	$PA_5 = \frac{5^1 + 2^1}{2^5 - 3^2} = \frac{7}{7} = 1$	$\mathbb{N}_{(+,5,1)}$	0.58%
		13	$\dots \rightarrow 13 \rightarrow 33 \rightarrow 83 \rightarrow 13 \rightarrow \dots$	$PA_5 = \frac{5^2 + 2^1 \cdot 5^1 + 2^2}{2^7 - 5^3} = \frac{39}{3} = 13$	$\mathbb{N}_{(+,5,13)}$	0.86%
		17	$\dots \rightarrow 17 \rightarrow 43 \rightarrow 27 \rightarrow 17 \rightarrow \dots$	$PA_5 = \frac{5^2 + 2^1 \cdot 5^1 + 2^4}{2^7 - 5^3} = \frac{51}{3} = 17$	$\mathbb{N}_{(+,5,17)}$	5.79%
		$PA_{ps} = 7$	$\dots \rightarrow \infty \rightarrow \dots$	—	$\mathbb{N}_{(+,5,\infty)}$	92.77%
181	C_{181,q_0}^{+}	1	$\dots \rightarrow 1 \rightarrow \infty$	—	$\mathbb{N}_{(+,181,\infty)}$	—
		27	$\dots \rightarrow 27 \rightarrow 611 \rightarrow 27 \rightarrow \dots$	$PA_{181} = \frac{181 + 2^5}{2^{15} - 181^2} = \frac{181}{27}$	$\mathbb{N}_{(+,181,27)}$	—
		35	$\dots \rightarrow 35 \rightarrow 99 \rightarrow 35 \rightarrow \dots$	$PA_{181} = \frac{181 + 2^6}{2^{15} - 181^2} = \frac{181}{35}$	$\mathbb{N}_{(+,181,35)}$	—
	C_{181,q_0}^{-}	1	$\dots \rightarrow 1 \rightarrow \infty$	—	$\mathbb{N}_{(-,181,\infty)}$	—

Table 2.6. Characteristics of the generalized transformations $C_{\kappa,q}^{\pm}$ with closed and open loops.

For these sequences, the trajectories are associated with the pseudo-attractor $PA_{ps} = 7$, corresponding to the node of the tree in Figure 2.3 generated by transformation (2.5) from the root sequence.

We see that as $n \rightarrow +\infty$ toward this point $PA_{ps} = 7$, the trajectories $CS_{5,q}^{+}$ of the transformations of the initial doubled numbers $q = 7, 11, 15, \dots$ converge.

However, the cycle is not closed, and therefore the point $PA_{ps} = 7$ serves as a pseudo-attractor.

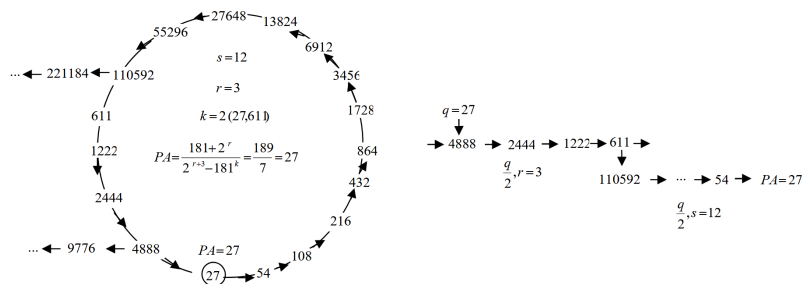


Figure 2.4. Examples of closed cycles with attractor $PA = 27$ corresponding to Collatz-type sequences.

2.9 Closed cycles and branching nodes

Property 2.7. *The number of branching nodes of the tree equals the number of odd numbers contained in a closed cycle with attractor PA.*

Proof. Consider a closed cycle of the transformation $C_{k,q}^{\pm}$ containing the odd values

$$q_0, q_1, \dots, q_{r-1}, \quad q_r = q_0.$$

Between two consecutive odd values q_i and q_{i+1} , the iteration consists of one transformation

$$q \mapsto \kappa q \pm 1,$$

followed by a finite number $s_i \geq 1$ of divisions by 2. Hence,

$$q_{i+1} = \frac{\kappa q_i \pm 1}{2^{s_i}}.$$

Equivalently, in the inverse Jacobsthal construction,

$$q_i = \frac{q_{i+1} 2^{s_i} \mp 1}{\kappa}.$$

Thus, each transition from one odd value of the cycle to the next corresponds to one branching node in the associated Jacobsthal tree.

Since the cycle contains exactly r odd values, and since each of these values generates exactly one inverse transition before the cycle closes, the corresponding Jacobsthal tree contains exactly r branching nodes along the closed cycle.

☐

3. Conclusion

We have developed a structural framework for generalized $\kappa x \pm 1$ transformations based on Jacobsthal sequences and their associated branching structures, which we call Jacobsthal trees. Within this representation, classical Collatz-type iterations appear as reverse trajectories along the branches of these trees. The periodicity of nodes, the recurrence relations governing their formation, and the structural identities derived in this work provide a unified description of the dynamics of $\kappa x \pm 1$ transformations. In particular, for a fixed κ , the natural numbers can be organized into disjoint subsets associated with distinct attractors or pseudo-attractors. Each Jacobsthal tree generates a set $N_{\kappa,PA}$ of odd nodes together with the corresponding even numbers $N_{\kappa,E}$, and these subsets appear to be mutually disjoint from the observed computational structures.

This motivates the following structural decomposition:

$$\mathbb{N} = \bigcup_{i=1}^v N_{\kappa, PA_i} \cup \bigcup_{i=1}^v N_{\kappa, E_i}.$$

Computational experiments performed on the first million initial values support this structural decomposition and confirm the observed non-intersection of subsets associated with different attractors.

The Jacobsthal-tree representation therefore provides an alternative structural viewpoint on generalized $\kappa x \pm 1$ discrete dynamical systems, highlighting intrinsic periodic symmetries and partition properties of the set $\mathbb{N}$.

Article Information

Author Contributions: Conceptualization: PK, DM; Methodology: PK, DM; Validation: PK, DM; Formal analysis: PK, DM; Investigation: PK, DM; Writing – original draft: PK, DM; Writing – review & editing: PK, DM; Project administration: DM.

Artificial Intelligence Statement: All mathematical results, definitions, proofs, computations, and scientific interpretations presented in this work were independently developed and validated by the authors. AI tools were used for language improvement, grammar correction, formatting assistance, and manuscript preparation. The authors take full responsibility for the content of this article.

Conflict of Interest Disclosure: The authors declare that they have no conflict of interest.

Plagiarism Statement: This article was scanned by the plagiarism program.

References

- [1] D. Barina, *Improved verification limit for the convergence of the Collatz conjecture*, J. Supercomput., **81** (2025), Article ID 810. <https://doi.org/10.1007/s11227-025-07337-0>
- [2] F. Kaplan, A. Özkoç Öztürk, *Binomial transforms of k -Narayana sequences and some properties*, Fundam. J. Math. Appl., **7**(3) (2024), 137–146. <https://doi.org/10.33401/fujma.1468536>
- [3] S. Uygun, *Binomial transforms of k -Jacobsthal sequences*, J. Math. Comput. Sci., **7**(6) (2017), 1100–1114. <https://doi.org/10.28919/jmcs/3474>
- [4] E. E. Kara, M. Ilkhan, *Some properties of generalized Fibonacci sequence spaces*, Linear Multilinear Algebra, **64**(11) (2016), 2208–2223. <https://doi.org/10.1080/03081087.2016.1145626>
- [5] E. Olgac, *Topology and structure of directed acyclic graphs*, in: Future of Information and Communication Conference (FICC 2021), Adv. Intell. Syst. Comput., **1364** (2021). https://doi.org/10.1007/978-3-030-73103-8_23
- [6] D. Bhattacharjee, *2-adic finite-certificate descent closure for the $3x+1$ Collatz problem*, Preprint (2025). Available online: https://www.researchgate.net/publication/405077396_2-Adic_Finite-Certificate_Descent_Closure_for_the_3x1_Collatz_Problem
- [7] M. B. Nathanson, *Permutation patterns of the iterated Syracuse function*, Integers, **24A** (2024), 1–24. Available online: <https://math.colgate.edu/~integers/vol24a.html>
- [8] D. Mailland, P. Kosobutskyy, *Modelling the Collatz problem from a Jacobsthal viewpoint*, CDS, **8**(1) (2026), 49–55. <https://doi.org/10.23939/cds2026.01.049>
- [9] P. Kosobutskyy, *The Jacobsthal-Collatz-Terras model of conjecture the natural numbers in $\kappa q + 1$ problems*, J. Appl. Math., **3**(2) (2025), Article ID 1767. <https://doi.org/10.59400/jam1767>
- [10] P. Kosobutskyy, O. Oborska, B. Vasylyshyn, *Jacobsthal recurrent numbers as a platform for transformations $\kappa q \pm 1$* , CDS, **7**(1) (2025), 288–299. <https://doi.org/10.23939/cds2025.01.288>
- [11] A. Horadam, *Jacobsthal representation numbers*, Fibonacci Quart., **34**(1) (1996), 40–54.
- [12] M. Pratsiovytyi, D. Karvatsky, *Jacobsthal–Lucas series and their applications*, Algebra Discrete Math., **24**(1) (2017), 169–180.
- [13] M. Chamberland, *An update on the $3x+1$ problem*, Working Paper, Department of Mathematics and Computer Science, Grinnell College (2003). Available online: https://chamberland.math.grinnell.edu/papers/3x_survey_eng.pdf
- [14] R. Terras, *A stopping time problem on the positive integers*, Acta Arith., **30**(3) (1976), 241–252.
- [15] P. Andolaro, *The $3x+1$ problem and directed graphs*, Fibonacci Quart., **40**(1) (2002), 43–54.
- [16] J. C. Lagarias, *The $3x+1$ problem: An overview*, In: J. C. Lagarias (Ed.), The Ultimate Challenge: The $3x+1$ Problem, Amer. Math. Soc., Providence, RI, 2010, 3–29.
- [17] S. Volkov, *A probabilistic model for the $5k+1$ problem and related maps*, Stochastic Process. Appl., **116**(4) (2006), 662–674. <https://doi.org/10.1016/j.spa.2005.11.007>
- [18] R. Crandall, *On the $3x+1$ problem*, Math. Comp., **32**(144) (1978), 1281–1292. <https://doi.org/10.2307/2006353>

- [19] C. Sultanov, C. Koch, S. Cox, *Collatz sequences in the light of graph theory*, Preprint, Potsdam University Repository (2020). <https://doi.org/10.25932/publishup-48214>
- [20] B. Snapp, M. Trac, *The Collatz problem and analogues*, J. Integer Seq., **11** (2008), Article ID 08.4.7.
- [21] L. Green, *The negative Collatz sequence*, version 1.25, 14 Aug. 2022. Available online: https://aplusclick.org/pdf/neg_collatz.pdf